



BRIEF CASES

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TEACHING NOTE

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Ceres Gardening Company: Funding Growth in Organic Products

Special Note to Instructors:

Two Excel spreadsheets are available to you from Harvard Business Publishing.

- Product 4019 presents **Case Exhibits 2** (Balance Sheet) and **3** (Income Statement) for students to use for forecasting financial performance. Included in the spreadsheet is a derived statement of cash flows, which underscores the cash effect of rapid growth in accounts receivable and inventories. Students will need to make assumptions about performance beyond 2006 in order to analyze the likely challenges with cash flow and debt covenants.
- Product 4020 presents a more comprehensive view of financial history and projections. Projections for financial performance beyond 2006 are included, as are calculations for the effect on channel inventory, dealer sales, and debt covenants. These projections match the ones shown in the teaching note, though the file allows instructors to experiment with their own sets of assumptions.

Introduction

The Ceres Gardening Company case presents a story of aggressive, continuing company growth and tacitly challenges students to identify the signs of potentially grave financial danger. More specifically, the case focuses on the interrelation between marketing strategy and credit policy and their effect on the company's financing requirements. The case illustrates the potential consequences when effective controls are not present to manage the way in which marketing drives revenue growth.

Professor John H. McArthur and writer Sunru Yong prepared this teaching note for the sole purpose of aiding classroom instructors in the use of the case "Ceres Gardening Company: Funding Growth in Organic Products" (HBS no. 4017). It provides analysis and questions intended to deepen students' comprehension of business issues and to energize classroom discussion. Cases are developed solely as the basis for class discussion and are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

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Case Synopsis

The case is set at the end of 2006. The Ceres Gardening Company is a leading player in the organic gardening industry. As consumer interest in organic products has grown, Ceres's distribution has shifted from direct catalog sales to retail sales through independent nurseries and garden centers. (Thus far Ceres has steered clear of distribution via giant home centers and discount stores.) Believing that the company is positioned to gain share as the organic gardening market grows, management began to market itself much more aggressively. The challenge in expanding distribution is that the window of time during which seeds and seedlings can be sold is narrow, and the small dealers that carry organic gardening products are unable to finance much inventory. Ceres responds to this obstacle with its GetCeres™ program, which offers aggressive discounts and very generous vendor financing. The objective is both to accelerate its penetration into new retail accounts and to encourage dealers to accept more inventory in anticipation of seasonal sales.

The decision ostensibly faced by Ceres's management is how to craft the right marketing plan for the following year, given the apparent success of the GetCeres™ program. What the company has failed to recognize is that significant inventory remains with the dealers, the accounts receivable has grown dramatically, and a likely decline in sales in the next year would put Ceres at risk of violating its debt covenants.

It is worth emphasizing that the case in no way suggests that the Ceres management was dishonest. Indeed, the actual protagonist of this heavily disguised case had good intentions and worked hard to create shareholder value. Furthermore, the market rewarded the company's growth with steadily increasing P/E and EBITDA multiples. Ultimately, it was inexperience and the absence of a qualified CFO that led the company into difficult times.

Learning Objectives

1. To guide students to identify critical clues in financial statements. A quick analysis of revenue and profit growth is often not sufficient to determine whether a company may be in trouble. The Ceres case is not about financial shenanigans, and a cursory view of the income statement would indicate that the company is operationally strong. Insightful students will recognize that the key to the case lies in the balance sheet, the notes to the financial statements, and the cash flow statement (not provided in the case).

2. To develop student skills in financial projection based on sound business judgment. Students should project possible outcomes for Ceres using reasonable assumptions based on information provided in the case. Dealer inventories have grown as Ceres has inadvertently "stuffed the channel," and this has important implications for future sales. Furthermore, the cost structure and debt covenants are relatively tight and do not allow much room for deterioration in performance.

3. To enable students to explore the connection between marketing and credit policy. Marketing and credit can—and often must—be complementary. Credit terms, applied judiciously, can be a useful marketing tool to win new accounts and gain greater share in existing accounts. However, it is a tool that can easily be misused, particularly in the hands of an ambitious sales and marketing department.

Secondary pedagogical goals may include discussion of organizational behavior in a growing firm and the issue of distribution channel strategy. Ceres is a company that has seen tremendous success in recent years. With top-line sales growing rapidly, it is very much a marketing-driven organization, as evidenced by the hiring of Annette O'Connell and how it has prioritized its resources. Indeed, in

the case—as well as at the actual company—the CFO role had not been adequately filled. The challenge of a changing distribution strategy is also a useful topic to raise. In most industries, neither sales nor profits remain static as customers evolve in how and where they purchase. This has important effects on everything from margins to advertising to sales force design.

Suggested Preparation Questions

The case can be taught in one of two ways. The company's predicament can emerge in the case discussion as Ceres management discovered it—namely, as a surprise (see *Approach 1: Management Perspective* below). This works best when students are not told the exact analysis required. We favor this approach, as it allows students to discover for themselves the difficulties confronting Ceres; the lesson that “cash is king” will make a deeper impression. Alternatively, the company's challenges can be identified through a directive analytical exercise (see *Approach 2: Financial Analysis Exercise* below). With this approach, students should be asked specifically to prepare a financial forecast. The optional student Excel file noted on page 1 of this teaching note should prove to be a useful tool.

Approach 1: Management Perspective

By design, the problems that Ceres faces are not obvious. The realization that Ceres is in trouble has greater impact if it emerges during the case discussion just as it did for the management of the disguised company—as a surprise. With this approach it usually takes the better part of an 80-minute class session for the students to realize these difficulties.

Bearing this in mind, the following preparation questions are intended to help students prepare for discussion, but not necessarily direct them to the answer:

1. What have been the key factors in the company's growth?
2. How has the stock market responded to the company's performance? Why?
3. What should Ceres's strategic plan be, given the trends in the organic gardening market?
4. How would you evaluate Ceres's marketing efforts? Should the GetCeres™ program be expanded? Why or why not?

Instructors who prefer to use somewhat more directive questions may consider the following. Additionally, providing the optional Excel file can help students forecast the cash flow challenge. However, the guidelines in this teaching note assume that most students will not have “cracked the case” prior to the class:

1. What potential financial risks does Ceres face as it crafts its strategy for 2007?
2. If you were the CFO, how would you approach the issue of crafting a marketing plan for 2007?
3. Prepare a presentation to the bank, including cash flow projections. What are the prospects for Ceres in 2007 and 2008? What risks might the company face?

Approach 2: Financial Analysis Exercise

The student Excel file will be a useful tool for students to calculate ratios and forecast financial performance in 2007 and beyond.

1. How has the company grown? What is its basic strategy and how has this evolved?

2. Is Ceres financially healthy? What parts of the financial statements might be a reason for confidence or a cause of concern?
3. Prepare a presentation to the bank. How would you forecast cash flow in 2007–2008? How might the sell-through of products affect this outlook?

Case Discussion and Analysis

The following guidelines to sequencing the case discussion assumes that the instructor will slowly lead students to the discovery of Ceres's dilemma. Each question represents a topic of discussion, along with suggested time allotment in an 80-minute class.

How has Ceres built its business? What decisions have been behind its success? (~10 minutes)

Students should recognize that Ceres was a relatively early mover in organic gardening. The company built a solid direct business when the industry was in its nascent stages, and developed a loyal following. On the supply chain side, a key component of growth was the network of organic supplier farms that Ceres developed. There were other organic seed catalog companies competing for the same consumer, but Ceres developed a platform that could be scaled up with the company.

In more recent years, the company recognized that the opportunity to grow revenues was shifting away from the direct model to the retail channel. Ceres remained fairly nimble, adapting its strategy accordingly, expanding its supplier network to better serve dealers outside its core geography, and bringing in a seasoned marketing veteran in Annette O'Connell.

What is Ceres's marketing strategy? What role has credit policy played in Ceres's growth? (~10 minutes)

The key elements of Ceres's marketing strategy were the following:

- Development of a full line of products, including seedlings and live plants that are appropriate to more casual gardeners and an increasingly wide selection of products
- An expanded operational infrastructure and sales team to support growth to drive rapid growth from local/regional to national market
- Market penetration ahead of competitors, including both organic seed companies and major conventional seed firms.

Credit policy was a critical driver of the marketing strategy, in light of two conditions. First, Ceres needed to put adequate inventory in retailers' hands *when needed by consumers*, given the seasonal nature of sales. Second, its customer base was primarily small, independent dealers unable to finance sufficient inventory. Providing generous vendor terms was standard industry practice, so there was a minimum threshold that Ceres had to meet. The GetCeres™ program upped the ante, making it far easier for cash-constrained, risk-averse dealers to bring in greater depth and breadth of Ceres inventory.

How well has Ceres done, and how should its performance be measured? (~10 minutes)

Students should see from the income statement that Ceres has grown profitably over the last several years, as shown in the compound annual growth rate. Profitability has improved, with EBIT and net income growing faster than sales. Furthermore, the annual growth rate has increased significantly in the last two years, since the inception of the GetCeres™ program:

CAGR	02-06E	04-06E
Revenue	15%	21%
EBIT	16%	12%

The growth of Ceres is particularly impressive when compared to the estimates in the industry report of **Case Exhibit 1**, which shows compound annual growth of 8.2% from 2002 to 2006. **Case Exhibit 5** shows that investors are happy with Ceres's performance. The regular dividend payment and growing sales have been recognized with the stock trading at an expanding multiple.

Stock Price Performance	2002	2003	2004	2005	2006E
Share Price, End of Year	\$ 4.11	\$ 6.21	\$ 6.41	\$ 8.01	\$ 9.49
Shares Outstanding (Millions)	3.0	3.0	3.0	3.0	3.0
Dividends / Share	\$ -	\$ 0.08	\$ 0.07	\$ 0.10	\$ 0.10
Dividend Yield	0.0%	1.2%	1.2%	1.2%	1.1%
EPS	\$ 0.40	\$ 0.43	\$ 0.43	\$ 0.50	\$ 0.51
P/E Ratio	10.4	14.4	15.0	16.2	18.6
EBITDA Multiple	8.00x	8.50x	8.90x	9.40x	10.20x

What are the risks faced by Ceres? (~10 minutes)

The discussion of risks for Ceres can be broadly grouped into the following categories:

Market demand uncertainty: Projected growth rates of organic gardening could change if consumer demand shifts. Trends may have been misunderstood, and some students may note that the projected market growth in **Case Exhibit 1** comes from an industry association, which may not be entirely unbiased. As with other outdoor-related spending, such as swimming pool installations, this industry is also vulnerable to regional weather issues. A longer winter in parts of the country, for example, can shorten the selling season and, at the margin, cause consumers to reduce purchasing.

Increasing competition: The case mentions the growing interest among conventional seed companies in pursuing the organic seed and seedling market. With well-established brands and national distribution, Ceres's expansion among dealers could be hurt.

Channel shifts and conflict: While niche products such as organic seeds and seedlings have a place in the independent nursery or garden center, the dominance of "big box" home centers (e.g., Home Depot, Lowe's) is a significant threat. Casual gardeners may be content to purchase their products in these channels, which could stymie growth where Ceres has most of its distribution. Students may note that the pilot project with Menards could be the company's initial effort to establish a foothold in a larger chain. However, the risk involved in pursuing deals with larger home centers is alienating the core independent dealer base, which may value product exclusivity and attention from Ceres's sales team.

Financial red flags: At least some students will have recognized problems based on analysis of the financial statements. Even without a statement of cash flows, the income statement and balance sheet alone reveal worrisome trends. The income statement shows revenue growth but declining profitability. Similarly, the balance sheet does not look particularly healthy—accounts receivable have grown dramatically and debt has increased.

Financial Metrics	2002	2003	2004	2005	2006E
Growth & Profitability					
Sales Growth		8.7%	9.3%	19.8%	21.4%
Gross Margin	17.0%	19.0%	18.6%	18.5%	17.6%
SG&A, % of Sales	8.1%	8.0%	8.1%	8.2%	8.4%
EBITDA, % of Sales	8.1%	10.3%	9.8%	9.7%	8.7%
Return on Sales	4.8%	4.8%	4.4%	4.2%	3.6%
Working Capital					
Accounts Receivable (Days Sales)	51.6	60.0	85.0	107.0	124.0
Inventories (Days COGS)	55.1	47.0	49.0	42.0	40.0
Inventory Turns	6.6	7.8	7.4	8.7	9.1
Accounts Payable (Days COGS)	36.3	50.0	75.0	85.0	98.0
Net Working Capital (CA - CL)	\$ 4,930	\$ 5,416	\$ 6,416	\$ 8,345	\$ 10,199
Working Capital Turnover (Sales ÷ NWC)	5.0	4.9	4.6	4.2	4.2
Current Ratio (CA ÷ CL)	3.1	2.6	2.2	2.1	2.0
Quick Ratio ((CA - Inv) ÷ CL)	1.8	1.8	1.6	1.7	1.6
Capital Structure					
Total Debt / EBITDA	1.80x	1.73x	2.18x	2.32x	2.48x
Debt / Enterprise Value	22%	20%	25%	25%	24%

Current Ratio = Current Assets ÷ Current Liabilities

Quick Ratio = (Current Assets – Inventories) ÷ Current Liabilities

Enterprise Value = Market Capitalization, end of period + Total Debt, end of period

Some students may create a statement of cash flows using the income statement and balance sheet. This will make the problem even more apparent, as operating cash flows have declined every year since the GetCeres™ program began.

Derived Statement of Cash Flows For Years Ending December 31	2003	2004	2005	2006E
Net Income	\$ 1,293	\$ 1,279	\$ 1,488	\$ 1,534
Depreciation & Amortization	412	455	557	669
Change in Accounts Receivable	(920)	(2,416)	(3,465)	(4,185)
Change in Inventories	294	(406)	(90)	(556)
Change in Accounts Payable	939	1,926	1,761	2,765
Operating Cash Flow	2,019	838	250	226
Investment in PP&E	(835)	(734)	(1,215)	(1,398)
Investment in Other Assets	-	-	-	-
Investment in Land	(1,300)	(1,103)	-	-
Investing Cash Flow	(2,135)	(1,836)	(1,215)	(1,398)
Debt Issuance	1,494	1,850	2,128	2,006
Retirement of Debt	(315)	(352)	(525)	(730)
Dividends	(226)	(224)	(298)	(307)
Financing Cash Flow	953	1,274	1,306	969
Change in Cash	\$ 837	\$ 276	\$ 340	\$ (203)

Project Ceres' financial performance in 2007 – what do you believe will happen? (~20 minutes)

There are two key pieces of analysis required to fully understand Ceres's predicament: (1) sell-through performance and implications for future sales; (2) projection of future profitability and potential covenant violations.

Most students will not have done the calculation to quantify the effects of channel-stuffing on 2007 sales. However, some will pick up on the clue at the end of the case about estimated dealer inventories, which suggests that Ceres has inadvertently stuffed the channel.

The estimated ending dealer inventories of \$10 million in 2005 and \$23 million in 2006 suggests that \$13 million of reported sales have not sold through to the end-consumer. Working backwards from this and incorporating what is known about overall market growth reveals that the years beyond 2006 may be difficult. Under virtually any reasonable assumption, Ceres sales to dealers will decline drastically.

Assuming that dealer inventories at the end of 2005 were in line with what dealers actually sought to carry, the target ending inventories can be estimated for subsequent years. The analysis here is done with a 15% growth rate, which is actually a more optimistic scenario than the ~10% annual growth rate mentioned in the industry report (see **Exhibit 1**).

Target Dealer Inventories	2005	2006E	2007F	2008F	2009F
Target Ending Dealer Inventories	10,000	11,500	13,225	15,209	17,490
Projected Growth, Consumer Sales (assumed)		15%	15%	15%	15%

This approach suggests target ending inventories of approximately \$11.5 million and \$13.2 million in 2006 and 2007, respectively. The estimated sell-through figure to consumers for 2006 can now be calculated using the following inventory formula:

$$\text{Beginning Inventories} + \text{Ceres Dealer Sales} - \text{Dealer Sell-Through} = \text{Ending Inventories}$$

We can solve for 2006 Dealer Sell-Through using case facts: (1) 80% of Ceres revenues are to dealers; (2) dealers had \$22 million in ending inventories:

$$\text{Dealer Sell-Through} = \text{Beginning Inventories} + \text{Ceres Dealer Sales} - \text{Ending Inventories}$$

$$\text{Dealer Sell-Through} = \$10.0 \text{ million} + \$34.1 \text{ million} - \$22.0 \text{ million} = \$22.1 \text{ million}$$

Implied sell-through is just \$22.1 million, leaving dealers with \$22.0 million of inventories compared to a target level of just \$11.5 million.

2006 Sell-Through and Ending Inventories	2006E	2007F	2008F	2009F
Beginning Dealer Inventories	10,000	16,500	13,225	15,209
+ Ceres Sales to Dealers	34,078	22,114	31,182	35,859
% of Ceres Sales to Dealers, 2006	80%			
Total Dealer Inventories, Full Year	44,078	38,614	44,407	51,068
- Dealer Sell-Through (implied for 2006)	22,078	25,389	29,198	33,577
Projected Growth, Consumer Sales (Assumed)		15%	15%	15%
Ending Dealer Inventories	22,000	13,225	15,209	17,490
Less Write-Off	(5,500)			
Estimated Inventory Write-Offs	25%			
Ending Dealer Inventories, 2006	16,500			

One problem that is not explicit in the case is that seedlings (and seeds, to a lesser extent) have a finite shelf life. Organic seedlings that have not been sold during the season become unsellable. While bad debt has not been a problem historically or in the first year of the GetCeres™ program, there is now a clear risk for the company. Significant amounts of unsellable product would leave Ceres with both bad debt and many angry, distrustful dealers who are likely to turn to other vendors. If 25% of this product is written off as unsellable, that would leave dealers with \$16.5 million to start 2007—still significantly above the estimated target level of \$11.5 million.

The same inventory formula can now be applied for future years. Starting with the target ending inventories for dealers and assuming growth in consumer sales (15% in the example above), we can solve for potential Ceres sales to dealers:

$$\text{Ceres Dealer Sales} = \text{Ending Inventories} + \text{Dealer Sell-Through} - \text{Beginning Inventories}$$

The analysis demonstrates that Ceres sales to dealers will fall dramatically in 2007 and will not return to 2006 levels until 2009. This carries important implications for Ceres's financial viability, particularly with regard to its debt covenants. The projections on the next page are based on the following assumptions:

- ~\$8.5 million of revenues (20% of 2006 sales) are Direct (catalog and internet)
 - ▶ Direct market is not growing, so the amount is constant in subsequent years
- Dealer sales match the 15% estimated growth as shown above
 - ▶ 2007: \$22.1 million
 - ▶ 2008: \$31.2 million
 - ▶ 2009: \$35.9 million
- Gross margins hold steady at 17.5% and SG&A is ~8.2% of sales
- Ceres makes no additional investment in PP&E, apart from maintenance capital expenditures that exactly offset depreciation
- Ceres does not issue any additional debt, but pays down existing debt

Income Statement For Years Ending December 31		2002	2003	2004	2005	2006E	2007F	2008F	2009F
Sales		\$ 24,652	\$ 26,797	\$ 29,289	\$ 35,088	\$ 42,597	\$ 30,634	\$ 39,701	\$ 44,378
Cost of Goods Sold		20,461	21,706	23,841	28,597	35,100	25,273	32,753	36,612
Gross Profit		4,191	5,091	5,448	6,491	7,497	5,361	6,948	7,766
General & Administrative Expense		1,999	2,138	2,372	2,877	3,578	2,512	3,255	3,639
Research & Development		203	203	212	222	232	239	246	254
Depreciation & Amortization		347	412	455	557	669	669	669	669
Earnings before Interest & Taxes		1,641	2,338	2,408	2,836	3,018	1,941	2,777	3,205
Interest		187	349	440	547	658	712	687	625
Earnings before Taxes		1,454	1,989	1,968	2,289	2,360	1,229	2,091	2,580
Taxes		264	696	689	801	826	430	732	903
Net Income		1,191	1,293	1,279	1,488	1,534	799	1,359	1,677
Balance Sheet At December 31		2002	2003	2004	2005	2006E	2007F	2008F	2009F
Assets									
Cash		705	1,542	1,818	2,158	1,955	4,310	2,476	1,873
Accounts Receivable		3,485	4,405	6,821	10,286	14,471	10,407	13,487	15,076
Inventories		3,089	2,795	3,201	3,291	3,847	3,116	4,038	4,514
Current Assets		7,279	8,742	11,839	15,735	20,273	17,833	20,002	21,463
Plant, Property, & Equipment (net)		2,257	2,680	2,958	3,617	4,347	4,347	4,347	4,347
Other Assets		645	645	645	645	645	645	645	645
Land		450	1,750	2,853	2,853	2,853	2,853	2,853	2,853
Non-Current Assets		3,352	5,075	6,456	7,115	7,844	7,844	7,844	7,844
Total Assets		10,631	13,817	18,295	22,850	28,117	25,678	27,846	29,307
Liabilities & Shareholders Equity									
Accounts Payable		2,034	2,973	4,899	6,660	9,424	6,232	8,076	9,028
Current Portion of Long-term Debt		315	352	525	730	649	763	763	695
Current Liabilities		2,349	3,325	5,423	7,390	10,074	6,995	8,839	9,722
Long-Term Debt		3,258	4,400	5,726	7,123	8,480	8,480	7,717	6,954
Shareholders Equity		5,024	6,091	7,146	8,336	9,563	10,203	11,290	12,631
Total Liabilities & Shareholders Equity		\$ 10,631	\$ 13,817	\$ 18,295	\$ 22,850	\$ 28,117	\$ 25,678	\$ 27,846	\$ 29,307

Ceres is barely above its break even point of \$30 million, mentioned at the end of the case. The projections above are based on dealer sales growing at 15%, which assumes that Ceres selects better dealers or equips them with the training and tools to sell more effectively than the overall market. That optimistic scenario could be wrong, and there is little room for error.

Break even Analysis	2006E
Overhead	5,230
Gross Margin	17.5%
Break even Sales	29,885

Furthermore, it should immediately be clear that Ceres will violate its debt covenants in 2007, even if no new debt is added.

Debt Covenants	2006E	2007F	2008F	2009F
Long-Term Debt / EBITDA	2.48x	3.54x	2.46x	1.97x
EBITDA / (Interest + Lease Obligation)	2.27x	1.56x	2.09x	2.44x

With the reduction in EBITDA in 2007, the long-term debt/EBITDA ratio climbs above the 3.25 maximum ratio. Similarly, the EBITDA / Fixed Claims ratio falls below 2.0.

What should Ceres do? (~20 minutes)

Given that the problem itself is intended to be something of a surprise, the case does not delve into the options available once Ceres understands its predicament. The issues that students should discuss as part of a solution may include the following:

Immediate

- *Renegotiation with lenders*—The fundamental business is sound, despite the misguided credit policies. The lenders do not face attractive options once the covenants have been violated, and Ceres can try to prove its capacity to cover its obligations under a restructured agreement.
- *Renegotiate with dealers*—Ceres may have significantly hurt its relationships with its dealers by “stuffing the channel.” It is likely that some of the product cannot be sold, and some dealers will be unable to pay for the inventory they accepted. Ceres may need to identify its most important accounts and look for ways to settle the outstanding receivables that will allow the company to maintain its dealer relationships for the long term.

Longer-Term

- *Reexamine decision-making process*—Marketing and sales teams tend to be aggressive by design. The steps that Ceres took into this problem were largely driven by a single department. It is not clear from the case that sufficient perspective is provided by anyone other than Marketing.
- *Hire the right people*—In the actual company on which the case is based, there was not a qualified CFO keeping an eye on the financial implications of its credit policy. This is not made evident in the case, but students should certainly question why this problem—rather easily identifiable—was completely overlooked. Part of the answer lay in not having the right people on board.